Laboratory 01: Signal Flow Analysis

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Introduction

Communication systems play an important role in modern telecommunications by allowing information to be transmitted between devices and users. These systems support many forms of communication, including voice calls, data transmission, video streaming, and internet services. Understanding how signals move through a communication system is essential for studying emerging technologies such as 5G, 6G, and Open Radio Access Networks (ORAN).

In this laboratory, I examined the basic flow of information through a communication system. Using Python and Matplotlib, I created a simple simulation that demonstrates how bits are transmitted, modulated, affected by channel noise, and recovered at the receiver. The purpose of this lab was to gain a better understanding of signal flow, BPSK modulation, and the effects of noise on communication systems.

Cell 1: Import Required Libraries and Create Original Bit Stream

```
import numpy as np
import matplotlib.pyplot as plt

bits = np.array([1, 0, 1, 1, 0, 0, 1, 0])
print("Original Bits:", bits)
```

```
Original Bits: [1 0 1 1 0 0 1 0]
```

Cell 2: Perform BPSK Modulation

```
modulated_signal = 2 * bits - 1
print("Modulated Signal:", modulated_signal)
```

```
Modulated Signal: [ 1 -1  1  1 -1 -1  1 -1]
```

Cell 3: Add Noise to the Channel

```
noise = np.random.normal(0, 0.4, len(modulated_signal))
received_signal = modulated_signal + noise

print("Received Signal with Noise:", received_signal)
```

```
Received Signal with Noise: [ 1.07377183 -0.6707351   1.96925389  1.34042928 -0.
  0.84498319 -0.57239357]
```

Cell 4: Recover the Bits (Demodulation)

```
recovered_bits = (received_signal > 0).astype(int)

print("Recovered Bits:", recovered_bits)
```

```
Recovered Bits: [1 0 1 1 0 0 1 0]
```

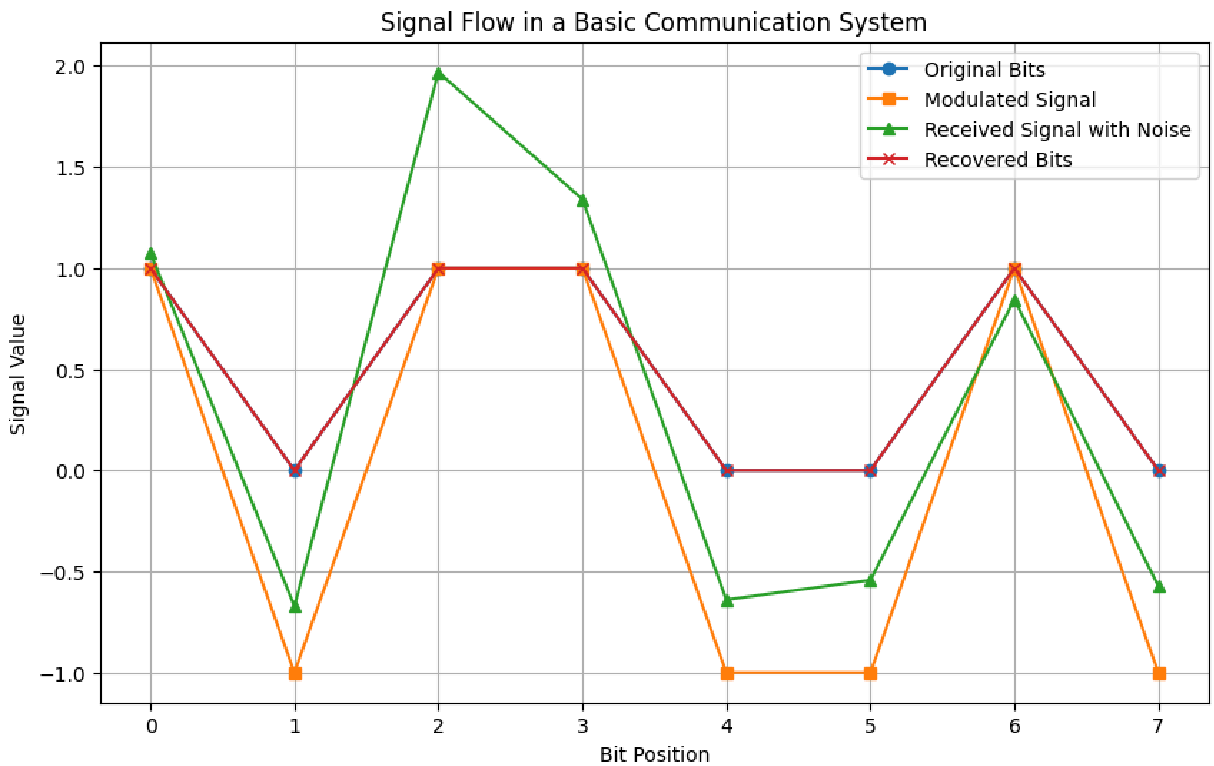
Cell 5: Plot Signal Flow Using Matplotlib

```
plt.figure(figsize=(10, 6))

plt.plot(bits, marker='o', label='Original Bits')
plt.plot(modulated_signal, marker='s', label='Modulated Signal')
plt.plot(received_signal, marker='^', label='Received Signal with Noise')
plt.plot(recovered_bits, marker='x', label='Recovered Bits')

plt.title("Signal Flow in a Basic Communication System")
plt.xlabel("Bit Position")
plt.ylabel("Signal Value")
plt.legend()
plt.grid(True)

plt.show()
```



Signal Flow Analysis Explanation

In this part of the lab, I used Python to show how a basic communication system works. The original bits were created first, then BPSK modulation changed the bits into signal values. A 1 became +1, and a 0 became -1. After that, noise was added to represent the channel. The demodulator then recovered the bits by checking whether each received signal value was greater than 0 or less than 0. The recovered bits matched the original bits, which shows that the signal was transmitted successfully even with some noise added.

Conclusion

In this laboratory, I learned how a basic communication system transmits information from a source to a receiver. Using Python and Matplotlib, I simulated the process of creating a bit stream, applying BPSK modulation, adding channel noise, and recovering the original bits through demodulation. The results showed that the communication system was able to successfully recover the transmitted information even when noise was present. This lab helped me better understand signal flow, modulation techniques, and the importance of communication systems in modern telecommunications and future 5G/6G networks.